

VeriPromise

Using in-context learning with Qwen3.5

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Task introduction

- 1. Promise Identification (PI):
 - Determine whether a segment expresses promising contents.
- 2. Supporting Evidence Assessment: assess
 - whether promises contain concrete evidence.
- 3. Clarity of the Promise–Evidence Pair
 - (CPEP): evaluate the clarity and relevance of
 - evidence in relation to the promise.
- 4. Timing for Verification (TV): indicate when
 - a promise should be revisited for verification
 - (e.g., within_two_years, two_to_five_years,
 - more_than_5_years, others).

子任務一：承諾識別

評估指標

- **Macro-F1**：精確率與召回率的調和平均
- 衡量模型識別 ESG 承諾語句的能力

子任務二：證據支持判斷

評估指標

- **Macro-F1**：判斷承諾是否具備充分支持證據
- 核心實務能力評估

子任務三：清晰度評估

評估指標

- **Macro-F1**：三分類（清楚/不清楚/誤導）平均表現
- 最具挑戰性任務，辨識漂綠風險能力

子任務四：驗證時機預測

評估指標

- **Macro-F1**：四分類時間推論能力
- 評估模型對承諾時間軸的理解

Dataset

```
{
  "id": 10002,
  "data": "在與供應商攜手邁向永續發展的過程中，台泥致力於建立一套以合作為基礎的價值體體系。核心原則是推動供應鏈的低碳轉型，同時維護人",
  "esg_type": "E",
  "promise_status": "Yes",
  "promise_string": "台泥致力於建立一套以合作為基礎的價值體體系。",
  "verification_timeline": "between_2_and_5_years",
  "evidence_status": "Yes",
  "evidence_string": "核心原則是推動供應鏈的低碳轉型，同時維護人權、保護環境和促進生物多樣性，以建立永續的合作夥伴關係。",
  "evidence_quality": "Not Clear",
  "company": "tcc",
  "ticker": 1101,
  "page_number": 32,
  "pdf_url": "https://hsxn1sjvkgtdpixe.public.blob.vercel-storage.com/tcc_1101_page_32.pdf",
  "company_source": "https://www.tccgroupholdings.com/tw/esgReport.html"
},
```

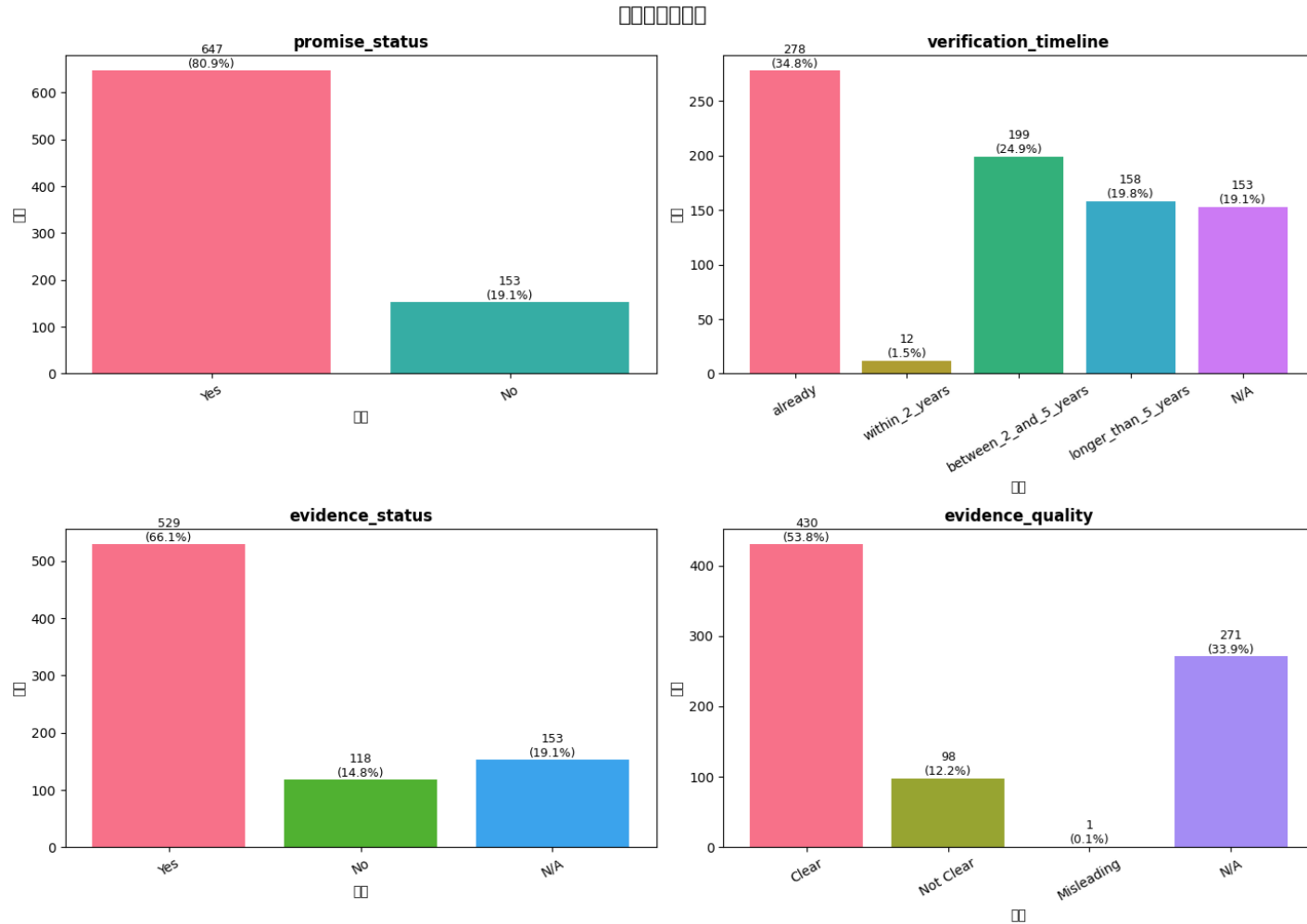
ESG type

- 環境 (Environmental, E)：關注企業對環境的影響，如溫室氣體排放（碳足跡）、能源效率、廢棄物管理、水資源管理、生物多樣性、產品綠化等。
- 社會 (Social, S)：關注企業與利害關係人的關係，如勞工權益、員工福利、工作健康與安全、多元平等（DEI）、數據隱私、產品責任、供應鏈社會責任。
- 公司治理 (Governance, G)：關注企業管理結構與倫理，如董事會組成、誠信經營、資訊透明度、風險管理、股東權利、供應鏈管理、商業道德。

Dataset

```
{
  "id": 10001,
  "data": "聯發科技除在「工作規則」中依照勞基法明確規定「員工在產假期間公司不得終止勞動契約」外，為支持同仁與其家人度過人生不同階段，自 2024 年起提供女性員工在分娩前後計有 12 週共 84 天的產假；男性員工則可於其配偶懷孕期間陪同產檢或生（流）產日及前後 15 日間請假陪伴，兩者合計共有 10 天陪產（檢）假可運用，陪產（檢）假期間工資照常給付。",
  "esg_type": "S",
  "promise_status": "Yes",
  "promise_string": "為支持同仁與其家人度過人生不同階段，自 2024 年起提供女性員工在分娩前後計有 12 週共 84 天的產假；男性員工則可於其配偶懷孕期間陪同產檢或生（流）產日及前後 15 日間請假陪伴，兩者合計共有 10 天陪產（檢）假可運用，陪產（檢）假期間工資照常給付。",
  "verification_timeline": "already",
  "evidence_status": "No",
  "evidence_string": "",
  "evidence_quality": "N/A",
  "company": "mediatek",
  "ticker": 2454,
  "page_number": 48,
  "pdf_url": "https://hsxn1sjvkgtdpixe.public.blob.vercel-storage.com/mediatek_2454_page_48.pdf",
  "company_source": "https://corp.mediatek.tw/about/sustainability"
},
```

Data distribution



Data Preprocess

- Remove additional form
 - "esg_type ", "company ", "ticker ", "page_number", "pdf_url", "company_source"
- Data spilit
 - Train:Test = 8:2

Load model

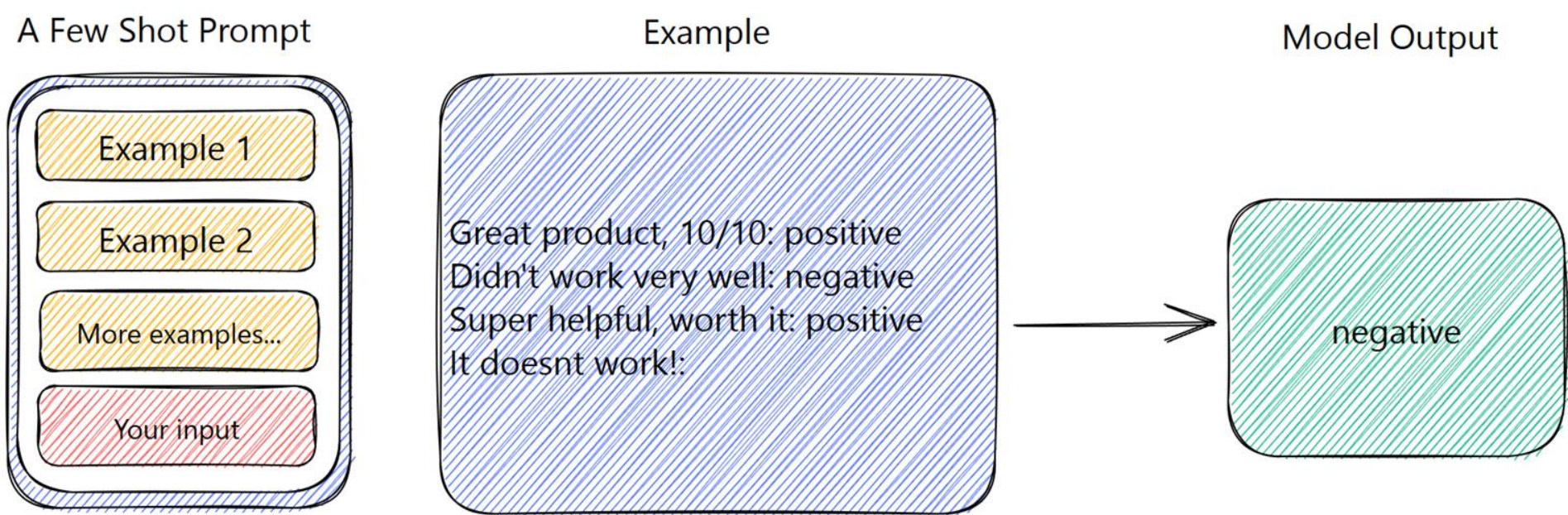
- Qwen/Qwen3.5-4B

- Type: Causal Language Model with Vision Encoder
- Training Stage: Pre-training & Post-training
- Language Model
 - Number of Parameters: 4B
 - Hidden Dimension: 2560
 - Token Embedding: 248320 (Padded)
 - Number of Layers: 32



- <https://huggingface.co/Qwen/Qwen3.5-4B>

Prompting



[Image source](#)

Prompting

Prompt 1: Field definitions:

1. `promise_status`: Whether the text contains an ESG promise/commitment. Values: Yes | No
2. `evidence_status`: Evaluate whether commitment statements are semantically clear and unambiguous, and Values: Yes | No | N/A
3. `evidence_quality`: Quality of the evidence provided.
Values: Clear | Not Clear | Misleading | N/A
4. `verification_timeline`: When the promise is expected to be fulfilled.
Values: already | within_2_years | between_2_and_5_years | longer_than_5_years | N/A

--- Examples ---

Text: 遠傳電信視連續兩年有交易之供應商為有效供應商，2024 年度有效供應商為 1,070 家，其中客戶裝置與資通訊類採購

`promise_status`: No

`evidence_status`: N/A

`evidence_quality`: N/A

`verification_timeline`: N/A

Text: 為減少各項能源耗用，合庫金控導入「ISO 50001 能源管理系統」，透過各項自發性之行動計畫與改善方案，致力節能、

`promise_status`: Yes

`evidence_status`: Yes

`evidence_quality`: Clear

`verification_timeline`: already

Predict

```
for idx, item in enumerate(test_set):  
    examples = rng.sample(train_set, FEW_SHOT_K)  
    prompt   = build_few_shot_prompt(examples, item)  
    output   = generate(prompt)  
    preds    = parse_output(output)
```

Prompt 1: {'promise_status': 'Yes', 'verification_timeline':
'already', 'evidence_status': 'Yes', 'evidence_quality':
'Clear'}

Scoring

最終分數 = 加權平均 Macro F1 :

- promise_status × 0.20
- evidence_status × 0.30
- evidence_quality × 0.35
- verification_timeline × 0.15

```
===== Evaluation Results =====  
promise_status           Macro F1 = 0.6893 (weight=0.2)  
verification_timeline    Macro F1 = 0.4243 (weight=0.15)  
evidence_status          Macro F1 = 0.5959 (weight=0.3)  
evidence_quality         Macro F1 = 0.3370 (weight=0.35)  
  
Weighted Macro F1 Score = 0.4982
```

Sample Code

- https://github.com/Bruceliangcf/veripromise_ICL

```
[(base) user@UserdeMini-37 Desktop % git clone https://github.com/Bruceliangcf/veripromise_ICL.git
Cloning into 'veripromise_ICL'...
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (17/17), done.
remote: Total 18 (delta 3), reused 11 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (18/18), 1.68 MiB | 2.90 MiB/s, done.
Resolving deltas: 100% (3/3), done.
[(base) user@UserdeMini-37 Desktop % conda create --name qwen3.5 python=3.11
```

Conda Enviroment

conda create —name <env name> python=3.11

```
Resolving deltas: 100% (3/3), done.
((base) user@UserdeMini-37 Desktop % conda create --name qwen3.5 python=3.11
2 channel Terms of Service accepted
Retrieving notices: done
Channels:
- defaults
Platform: osx-arm64
Collecting package metadata (repodata.json): done
Solving environment: done

==> WARNING: A newer version of conda exists. <==
  current version: 25.11.1
  latest version: 26.1.1

Please update conda by running

  $ conda update -n base -c defaults conda

## Package Plan ##

environment location: /opt/anaconda3/envs/qwen3.5

added / updated specs:
- python=3.11
```

Conda Enviroment

type “y” continually install

```
The following NEW packages will be INSTALLED:
```

bzip2	pkgs/main/osx-arm64::bzip2-1.0.8-h80987f9_6
ca-certificates	pkgs/main/osx-arm64::ca-certificates-2025.12.2-hca03da5_0
libcxx	pkgs/main/osx-arm64::libcxx-21.1.8-hb4ce287_0
libexpat	pkgs/main/osx-arm64::libexpat-2.7.4-h50f4ffc_0
libffi	pkgs/main/osx-arm64::libffi-3.4.4-hca03da5_1
libzlib	pkgs/main/osx-arm64::libzlib-1.3.1-h5f15de7_0
ncurses	pkgs/main/osx-arm64::ncurses-6.5-hee39554_0
openssl	pkgs/main/osx-arm64::openssl-3.5.5-ha0b305a_0
packaging	pkgs/main/osx-arm64::packaging-25.0-py311hca03da5_1
pip	pkgs/main/noarch::pip-26.0.1-pyhc872135_0
python	pkgs/main/osx-arm64::python-3.11.15-h4f1bc5c_0
readline	pkgs/main/osx-arm64::readline-8.3-h0b18652_0
setuptools	pkgs/main/osx-arm64::setuptools-80.10.2-py311hca03da5_0
sqlite	pkgs/main/osx-arm64::sqlite-3.51.2-h67002bf_0
tk	pkgs/main/osx-arm64::tk-8.6.15-hcd8a7d5_0
tzdata	pkgs/main/noarch::tzdata-2026a-he532380_0
wheel	pkgs/main/osx-arm64::wheel-0.46.3-py311hca03da5_0
xz	pkgs/main/osx-arm64::xz-5.8.2-h8bbcb1d_0
zlib	pkgs/main/osx-arm64::zlib-1.3.1-h5f15de7_0

```
Proceed ([y]/n)? #
```

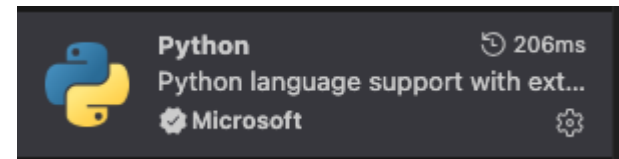
Conda Enviroment

switch to your conda environment.

```
[(base) user@UserdeMini-37 Desktop % conda activate qwen3.5  
(qwen3.5) user@UserdeMini-37 Desktop % █
```


VS CODE

Install some extensive module in Vs Code.

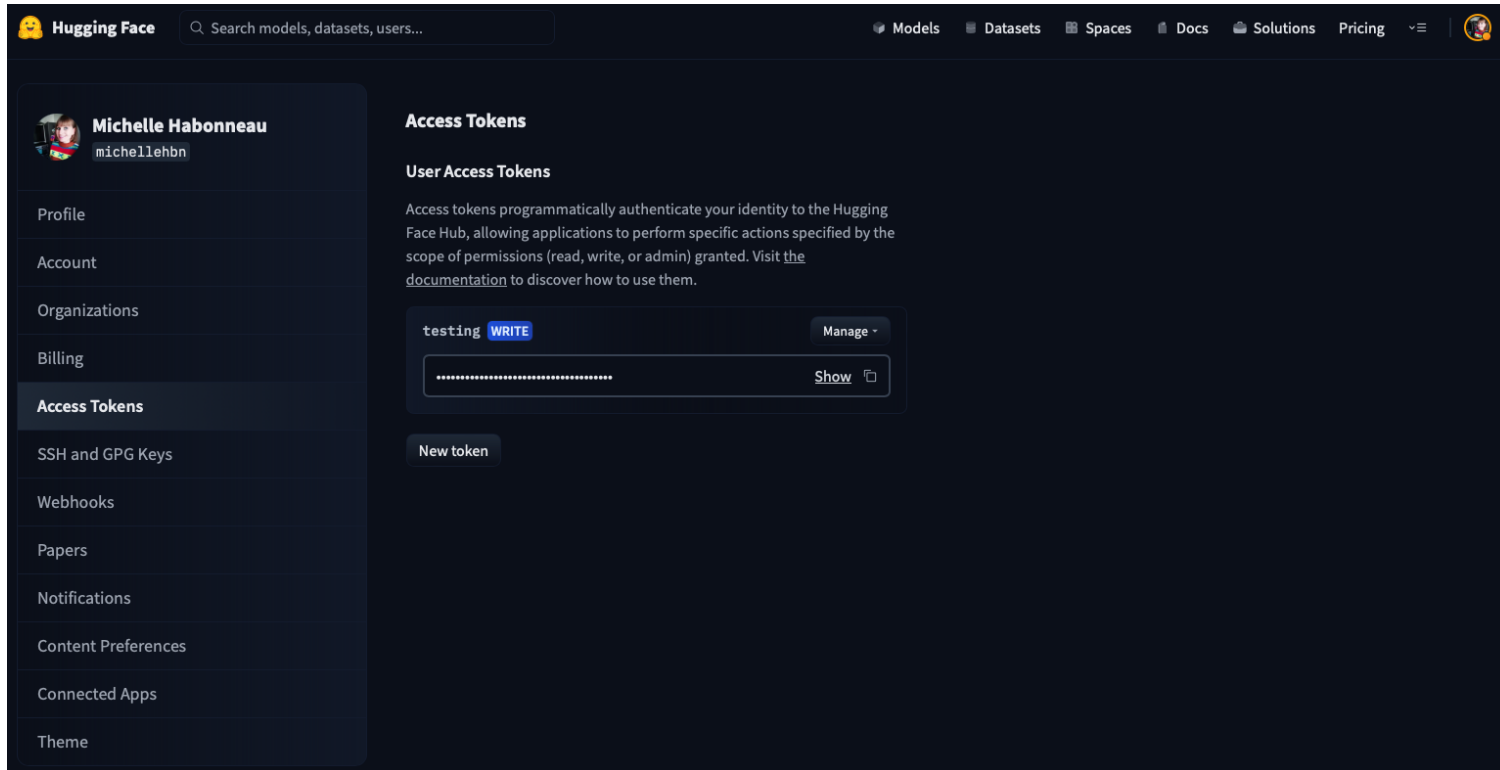


Install envirrroment

- `pip install ipykernel transformers torch torchvision scikit-learn pandas matplotlib seaborn accelerate`
- `pip install -U "huggingface_hub[cli]"`
- After install, you can use “`pip list`” to check package version.

```
[(base) user@UserdeMini-37 Desktop % git clone https://github.com/BruceLiangcf/veripromise_ICL.git
Cloning into 'veripromise_ICL'...
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (17/17), done.
remote: Total 18 (delta 3), reused 11 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (18/18), 1.68 MiB | 2.90 MiB/s, done.
Resolving deltas: 100% (3/3), done.
[(base) user@UserdeMini-37 Desktop % conda create --name qwen3.5 python=3.11
```

Huggingface Access token



<https://huggingface.co/docs/hub/security-tokens>

Huggingface Access token

hf auth login

hf auth whoami

hf auth logout

```
from transformers import AutoModelForCausalLM, AutoTokenizer
import torch

MODEL_NAME = "Qwen/Qwen3.5-0.8B"
print(f"Loading model: {MODEL_NAME} ...")

tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME, trust_remote_code=True)
model = AutoModelForCausalLM.from_pretrained(
    MODEL_NAME,
    torch_dtype=torch.float16,
    device_map="mps",
    trust_remote_code=True,
)

model.eval()
print("Model loaded.")
```

[10] ✓ 54.8s Python

... Loading model: Qwen/Qwen3.5-0.8B ...
The fast path is not available because one of the required library is not installed. Falling bac
Loading weights: 100%|██████████| 320/320 [00:00<00:00, 358.22it/s]
Model loaded.